

Homework 3

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Homework Deliverables

1 Step-by-step method (concise)

1.1 (i) Force / energy evaluation

We model the beam with discrete stretching and bending energies per node.

Stretching (between nodes $k, k + 1$):

$$E_s^k = \frac{1}{2}EA\ell_k \left(\frac{\|\mathbf{x}_{k+1} - \mathbf{x}_k\|}{\ell_k} - 1 \right)^2, \quad \mathbf{F}_s^k = -\nabla_{\mathbf{q}} E_s^k, \quad \mathbf{J}_s^k = -\nabla_{\mathbf{q}}^2 E_s^k.$$

Bending (nodes $k - 1, k, k + 1$):

$$E_b^k = \frac{1}{2} \frac{EI}{\ell_k} (\kappa_k - \bar{\kappa}_k)^2, \quad \kappa_k = \frac{2\mathbf{t}_e \times \mathbf{t}_f}{1 + \mathbf{t}_e \cdot \mathbf{t}_f},$$

with unit tangents $\mathbf{t}_e, \mathbf{t}_f$. Gradients/Hessians are computed analytically following Bergou et al.

Pseudocode (forces & Jacobian)

```
Function getElasticForces(q, EA, EI, deltaL):  
    initialize F_el, J_el (zeros)  
    for each segment k: compute grad_Es, hess_Es; accumulate into F_el, J_el  
    for each interior node k: compute grad_Eb, hess_Eb; accumulate  
    return F_el, J_el
```

1.2 (ii) Time stepping

We use a fully implicit (backward Euler) integration scheme with Newton–Raphson iteration to solve for the next configuration.

Discrete equation of motion:

$$\mathbf{M} \frac{\mathbf{q}^{n+1} - \mathbf{q}^n}{\Delta t^2} - \mathbf{M} \frac{\mathbf{u}^n}{\Delta t} = \mathbf{F}_{\text{elastic}}(\mathbf{q}^{n+1}) + \mathbf{W}.$$

Residual and Jacobian:

$$\mathbf{f}(\mathbf{q}) = \mathbf{M} \frac{\mathbf{q} - \mathbf{q}^n}{\Delta t^2} - \mathbf{M} \frac{\mathbf{u}^n}{\Delta t} - \mathbf{F}_{\text{elastic}}(\mathbf{q}) - \mathbf{W}, \quad \mathbf{J} = \frac{\mathbf{M}}{\Delta t^2} - \mathbf{J}_{\text{elastic}}.$$

Velocity update (after convergence):

$$\mathbf{u}^{n+1} = \frac{\mathbf{q}^{n+1} - \mathbf{q}^n}{\Delta t}.$$

Pseudocode (time step)

```
Function timeStep(q_old, u_old, dt, tol, max_iter, M, EA, EI, W, deltaL, free_DOF):
    q = q_old
    for iter in 1..max_iter:
        F_inertia = M/dt * ((q - q_old)/dt - u_old)
        J_inertia = M / dt^2
        F_el, J_el = getElasticForces(q, EA, EI, deltaL)
        f = F_inertia - F_el - W
        J = J_inertia - J_el
        solve J[free_DOF, free_DOF] * dq = f[free_DOF]
        q[free_DOF] -= dq
        if norm(f[free_DOF]) < tol: return q, success
    return q, failure
```

1.3 (iii) Dirichlet constraints

The left end (node 0) is clamped:

$$x_0 = 0, \quad y_0 = 0.$$

The right end is controlled:

$$(x_N, y_N) = (x_c, y_c), \quad (x_{N-1}, y_{N-1}) = (x_c, y_c) - \Delta L(\cos \theta_c, \sin \theta_c).$$

1.4 (iv) PI control law

We track the middle node (x_j, y_j) , where $j = \lfloor (N+1)/2 \rfloor$, using a PI controller that adjusts (x_c, y_c) .

Target trajectory (quarter circle, $T = 1000$ s):

$$x^*(t) = \frac{L}{2} \cos\left(\frac{\pi t}{2T}\right), \quad y^*(t) = -\frac{L}{2} \sin\left(\frac{\pi t}{2T}\right).$$

PI update:

$$E_x^{n+1} = E_x^n + e_x \Delta t, \quad E_y^{n+1} = E_y^n + e_y \Delta t,$$

$$x_c^{n+1} = x_c^n + K_{p,x} e_x + K_{i,x} E_x^{n+1}, \quad y_c^{n+1} = y_c^n + K_{p,y} e_y + K_{i,y} E_y^{n+1},$$

and

$$\theta_c = \text{atan2}(y_c - y_j, x_c - x_j).$$

2 Control Inputs Over Time

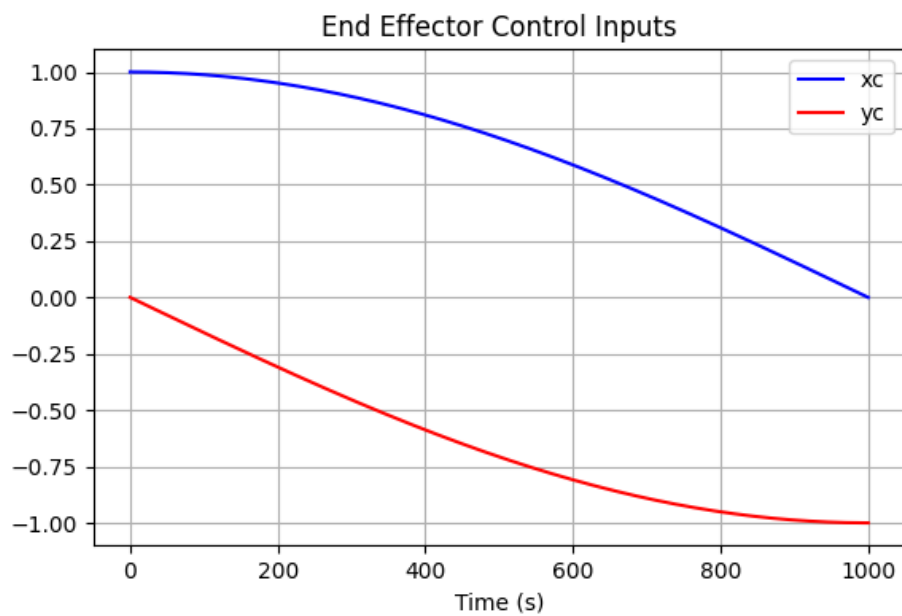


Figure 1: Shows the time evolution of the control inputs $x_c(t)$ and $y_c(t)$. The PID controller smoothly adjusts the end effector position to track the desired circular trajectory.

- $x_c(t)$ decreases from 1.0 m to approximately 0.5 m, following the cosine trajectory
- $y_c(t)$ decreases from 0 m to approximately -0.5 m, following the negative sine trajectory

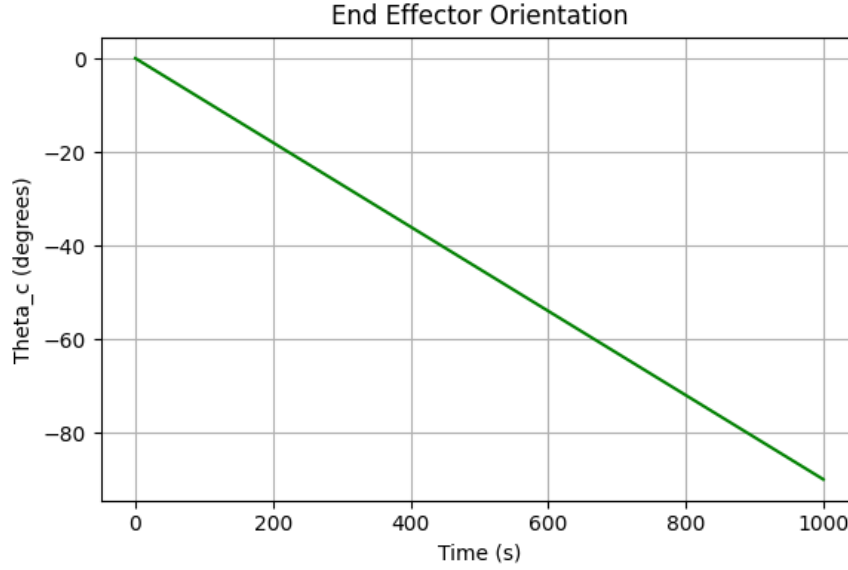


Figure 2: Shows the time evolution of the control input $\theta_c(t)$.

- $\theta_c(t)$ transitions from 0° to approximately -90° as the beam bends downward

3 Beam Shape Snapshots

Figures 3-7 show five snapshots of the beam configuration at different time instances during the motion. The beam successfully deforms from its initial straight horizontal configuration to follow the prescribed quarter-circle trajectory. The middle node (red marker) closely tracks the target position (shown in red), while the end effector (blue marker) is controlled to achieve this tracking.

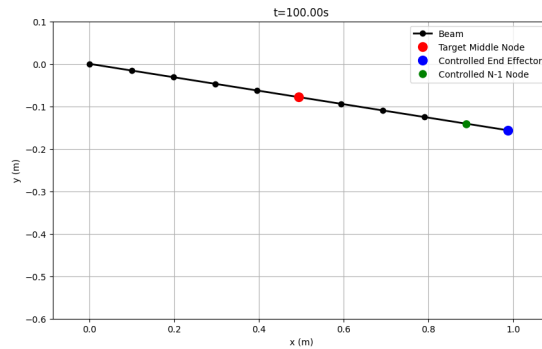


Figure 3: Beam begins to curve downward

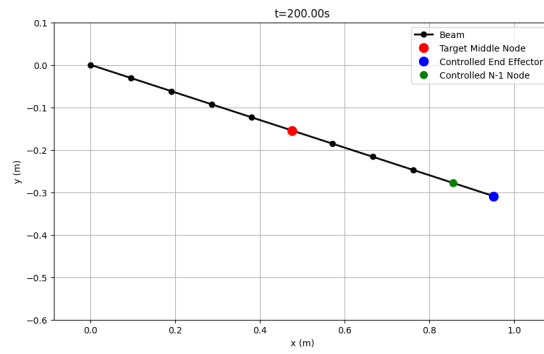


Figure 4: Beam continues to curve downward

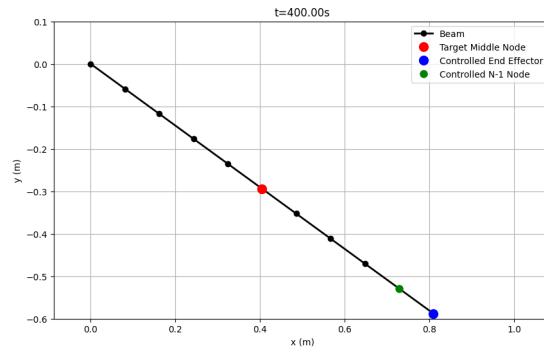


Figure 5: Beam continues to curve downward

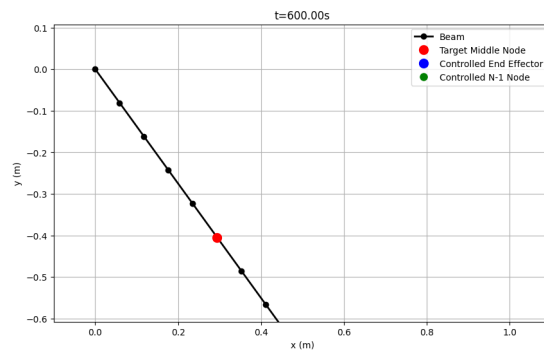


Figure 6: Beam continues to follow the arc.

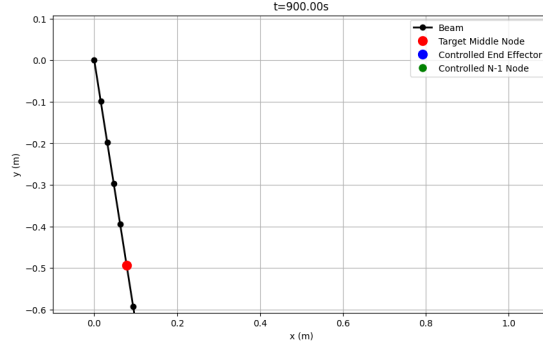


Figure 7: Beam almost at final configuration.

4 Feasibility Discussion

4.1 Workspace Limits

The robotic end effector is constrained to operate within a physical workspace defined by:

$$0 \leq x_c \leq L \quad (1)$$

$$-L \leq y_c \leq 0 \quad (2)$$

These constraints represent:

- **Horizontal limits:** The robot cannot extend beyond the beam length $L = 1$ m or retract below the origin
- **Vertical limits:** The robot operates below the horizontal plane (gravity acts downward) and cannot extend above $y = 0$

4.2 Handling Infeasible Commands

When the PID controller computes a control input that violates workspace constraints, we employ **saturation** (clamping):

$$\begin{aligned} x_c &= \max(0, \min(x_c, L)) \\ y_c &= \max(-L, \min(y_c, 0)) \end{aligned}$$

4.3 Joint Limits and Orientation Constraints

The orientation angle θ_c is geometrically constrained by the beam configuration. We compute it based on the actual middle node position:

$$\theta_c = \arctan 2(y_c - y_j, x_c - x_j) \quad (3)$$

This ensures that θ_c remains consistent with the physical geometry and avoids singularities.

4.4 Rate Limits

In practice, robotic actuators have maximum velocity and acceleration limits. Our implementation implicitly respects rate limits through:

- **Small time steps:** $\Delta t = 0.01$ s limits the maximum position change per step
- **Moderate gains:** $K_p = 0.15$ prevents excessive control action
- **Smooth trajectory:** The target trajectory has continuous derivatives, avoiding sudden changes

References

- [1] M. K. Jawed, “Lecture 5–Modules 10–12,” Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class hand-out].
- [2] M. K. Jawed, “Lecture 6–Modules 13–20,” Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class resource].
- [3] M. K. Jawed, “Simulation of a Beam with Fixed DOFs,” Dept. of Mechanical and Aerospace Engineering, Univ. of California, Los Angeles, 2025. [Class resource].